Two election-night models compared: NYC 2025 general vs. TX 2026 primary · Young-Kyung Kim · October 18, 2026

Both pipelines in `scripts/model/modeling-cory` forecast an outcome from partial precinct returns, both were run live, and they share an author and a utility package. Architecturally they have almost nothing else in common. This memo sets them side by side — outcome, likelihood, prior, estimation, uncertainty — and records what each one measurably did on the night. All diagnostics quoted here were computed from the committed code and its logged output; sources are given in §9.

1. What each system was

	NYC	TX
Contest	2025 NYC mayoral, *general*	2026 TX *primaries* (Senate, House)
Election day	4 November 2025	3 March 2026
Unit	election district (precinct)	precinct × vote method
Units modelled	4,061	6,079 (Senate GOP)
Candidates	3 (Mamdani, Cuomo, Sliwa)	up to 8 per contest
Live runs logged	24, 21:31–00:27 EST	21, 01:38–21:14
Estimator lives in	`enightmodels` (binary dep.)	the repo itself

One asymmetry is worth noting up front: TX's model is fully readable in the repository, whereas NYC's core is a compiled dependency (`CoryMcCartan/enightmodels`, pinned at `f3edb6e`) whose source tree is not checked in. The only copy on disk is the installed build, readable via `lazyLoad` on its database.

2. Outcome and likelihood

This is the deepest divergence, and most of the others follow from it.

NYC: Gaussian, turnout and share modelled separately. Two kinds of equation, sharing one control formula:

$$\log(\tau_i + 0.1) = x_i'\beta^\tau + \varepsilon_i^\tau, \qquad s_{ij} = x_i'\beta_j + \varepsilon_{ij}, \quad j = 1, \dots, 3. \tag{1}$$

The share equations are linear in the *level* of the share — no logit, no multinomial — and are fit independently. Predictions are clamped to $[0, 1]$, observed values substituted for reported precincts, and then normalised so the three shares sum to one. That division is the only thing coupling the candidates.

TX: Poisson on counts, turnout implicit. One model per candidate per method:

$$\log \mathbb{E}[v_{icm}] = \beta_0 + \beta_1 \log(\widehat{\text{est}}_{icm} + 0.1) + f(\text{covariates}) + u_{\text{stratum}} + u_{\text{geostratum}}. \tag{2}$$

Because counts are modelled, turnout is implied ($\tau_{im} = \sum_j v_{ijm}$) rather than estimated, and shares are formed afterwards as $(0.1 + \hat{v})/(0.1 + \sum_j \hat{v})$. Note $\log(\widehat{\text{est}})$ enters as a free-coefficient covariate, not a fixed offset, so the relationship to expected turnout is a fitted power law.

On this axis, TX is structurally closer to right

Modelling counts per candidate *per mode* is implicitly a joint turnout-and-share model, which matters because a precinct's overall share is a turnout-weighted mixture over modes, $s_{ij} = \sum_m s_{ijm}\tau_{im} / \sum_m \tau_{im}$. If mail splits one way and election day another, the precinct's share is determined by the *mix*. NYC can treat turnout and share as separable only because it drops mode entirely — it reads `vote_mode` and then discards it, absorbing the effect into covariates such as `prim25_cvr_eday` and an early-vote threshold. For a general election whose reporting order is mode-driven, that is the weaker choice.

3. The prior

NYC: a strong, informative g-prior. The distinctive feature of the whole pipeline. The prior is placed on *fitted values*, not coefficients, and implemented by stacking all N precincts as pseudo-observations with weight $w = n_0/N$ on top of the reported rows. The implied prior is exactly Zellner's g-prior,

$$\beta \sim \mathcal{N}\big((X'X)^{-1}X'\tilde{y},\ \sigma^2(wX'X)^{-1}\big), \tag{3}$$

with the location being the OLS projection of the prior fitted values. ("g-prior" names Zellner's structure, in which the prior covariance is proportional to $(X'X)^{-1}$; it is not shorthand for "Gaussian prior," and a ridge prior $\mathcal{N}(0, \lambda^{-1}I)$ is Gaussian but not a g-prior. The proportionality to $X'X$ is what makes n_0 readable as an effective sample size — and also why no choice of n_0 improves the conditioning of the design.) Those fitted values come from a vendor likely-voter file (per-precinct `pred.turnout`, `pred.mamdani`, `pred.cuomo`, `pred.sliwa`, as of 9 October 2025), rescaled by a single logit shift so that weighted aggregates hit exogenous targets: borough turnout totals, and citywide shares of 0.47/0.38/0.12.

TX: deliberately uninformative about candidates. There is no candidate-level prior at all. Each candidate's baseline is expected party turnout divided by the number of candidates,

$$\widehat{\text{est}}_{icm} = \text{est}_{im} \cdot \frac{\text{tr}}{n_{\text{cand}}}, \tag{4}$$

i.e. a flat split. Every candidate-level signal has to come from reported precincts. For a primary with no precinct-level polling this is the right call, and it is a genuine design difference rather than an omission.

Consequence: the two models move for different reasons. In NYC the prior's relative weight decays automatically as $n_0/(n_0 + n_{\text{obs}})$, so the forecast walks from the vendor prediction toward the data. In TX there is nothing to walk from: early-night behaviour rests entirely on $\log(\widehat{\text{est}})$ and a turnout-ratio adjustment.

4. Estimation

The Poisson log-likelihood $\sum_i \{v_i x_i'\beta - e^{x_i'\beta}\}$ is not quadratic in β, so no finite-dimensional conjugate family exists. (Gamma conjugacy applies to a free rate λ_i; imposing $\lambda_i = e^{x_i'\beta}$ with β shared across precincts destroys it.) TX must therefore optimise, which yields a point estimate and no posterior — and that single fact explains its uncertainty machinery, §5.

	NYC	TX
Inference	conjugate Bayes, closed form	penalised ML
Machinery	one weighted QR per equation	`lme4::glmer`, Laplace approx.
Fallback	—	`glmnet` Poisson, fixed $\lambda = 10$
Returns	full posterior	point estimate
Draws	500 per equation	none
Regularisation	n_0 pseudo-rows (hand-set)	random intercepts; lasso at $\lambda = 10$
Iterates?	no	yes (IRLS / coordinate descent)

The `glmnet` fallback has an uncontrolled prior

`glm_minnorm` calls `glmnet(X, y, family, lambda = 10)` with no `alpha`, so it is *lasso*. On a representative Poisson fit (Cornyn counts, 4,221 precincts, the model's own demographic covariates) $\lambda = 10$ zeroes five of eight coefficients — including `dem24_est`, the 2024 partisan baseline, and both race covariates — against $\lambda_{\max} = 73.5$ for that data. The structural problem is not the magnitude but that λ is a fixed absolute number while $\lambda_{\max}$ is data-dependent: effective prior strength then varies unpredictably across contests and across the night. At fifty-state scale this is unmanageable.

5. Uncertainty

NYC: posterior draws, two intervals. Coefficients are drawn as $\beta^{(d)} = \hat{\beta} + R^{-1}z^{(d)}$ with $z^{(d)} \sim \mathcal{N}(0, \sigma^{(d)2}I)$ and $\sigma^{(d)}$ from a scaled inverse-χ^2; predictive noise resamples leverage-corrected residuals $e_i/(1-h_i)$ once more than 25 precincts report. Aggregation is draw-wise, $S_j = \sum_i s_{ij}\tau_i / \sum_i \tau_i$, so turnout and share uncertainty propagate jointly. Both a credible interval and an anytime-valid confidence sequence are reported at level 0.9999.

TX: conformal residual quantiles. With no posterior, intervals are built from the cross-precinct spread of within-precinct share errors,

$$r_{ic} = \frac{v_{ic}}{\sum_{c'} v_{ic'}} - \frac{\hat{\mu}_{ic}}{\sum_{c'} \hat{\mu}_{ic'}}, \tag{5}$$

whose $\alpha/2$ and $1-\alpha/2$ quantiles across precincts are then added to *every* unreported precinct's share and summed.

The TX interval assumes perfectly correlated precinct errors

Adding the same quantile to all ~6,000 precincts and summing is algebraically the assumption that precinct-level errors move together as one common shock. A measure of *cross-sectional dispersion among* precincts is thereby reported as *uncertainty about their sum*. Under independence the aggregate error would scale as $\sigma_p/\sqrt{N}$, roughly 77× narrower; the truth lies between, but nowhere near 1×. Two further problems in the same block: the residuals are *in-sample* (predicted on the very precincts used to fit), so there is no calibration split and no coverage guarantee — hence the `approx_` label — and `conf_level` defaults to 0.95 while the output field is hardcoded `"approx_conform_99"`, so consumers are told 99% while 95% quantiles are computed.

The consequence is visible in the logs. For the Senate GOP contest the point estimate was stable all night (Cornyn 40.6–43.9%), but the interval *widened* from 46.4 to a peak of 63.5 percentage points as counting progressed, because $q_{\min}, q_{\max}$ are re-estimated each run from an increasingly heterogeneous conformity set and that growth swamps the narrowing from substituted precincts. At the final logged

run Cornyn's interval was [25.0, 73.2] around a point estimate of 43.1% — fully overlapping Paxton's [21.3, 69.8].

By contrast NYC's widths behave as theory predicts: the confidence sequence runs 1.33–1.45× the credible interval at the final snapshot, the premium one pays for validity under roughly a hundred looks.

6. Ensembles and specifications

	NYC	**TX**
Count	4 specs, all reported	7 specs in 4 fallback chains, + county legacy
Selection	none; consumers choose	`map_until` takes the first that does not error
Complexity	2 to 21 covariates	intercept-only to a 43-term spline
Prior strength	n_0 from 1.0 to 15.5, by eye	n/a
Nonlinearity	none, all linear	dem24_est^2, thin-plate spline
Spatial	64-term Fourier basis, *disabled*	`geostratum` effects, active
Partial pooling	none	`stratum_demo` × `geostratum` random effects

NYC leans on many hand-picked covariates plus a strong prior; TX on few covariates plus strata and splines with no candidate prior. For hundreds of races the TX approach is the one that generalises — one cannot hand-curate 21 covariates per contest — and its 5 × 6 strata cross is already the partial-pooling skeleton that sparse legislative districts would need.

7. Two measured problems specific to each

NYC: the prior is lossy in one spec, and silently so. Because the g-prior location is the projection of $\tilde{y}$ onto col(X), only the component the covariates can express survives. Measuring the centred R^2 of each vendor prediction on each spec's own column span:

spec	Mamdani	Cuomo	Sliwa
`simple_pres`	0.544	**0.073**	0.711
`simple_pred`, `simple_mult`	1.000	1.000	1.000

Turnout is always safe, since log(pred_turn) is a covariate everywhere. But `simple_pres` does not contain the vendor share predictions, so its prior on Cuomo's share is effectively a constant: 93% of the vendor's precinct-level Cuomo signal is annihilated by the projection. The four specs are therefore not a n_0 ladder over a common prior — they encode materially *different* priors, and since the prior predictive feeds the confidence sequence, that propagates into the reported bands.

TX: two mechanisms never engaged. The turnout-ratio adjustment filters on `stage >= 6`, but the precinct feed emits only stages 0, 2 and 5 (counts 20/1511/2776). The lookup table is therefore always empty and every run silently used the hardcoded `turn_ratio = 0.95`. Separately, `pct_rep` in the logs is non-monotonic — 0.997 → 0.269 → 0.364 — so the feed regressed and the model's reported base collapsed mid-count. Both are plumbing faults rather than modelling errors, and both are invisible without reading the logs.

8. Where each one's design should carry into 2026

Component	Take from	Rationale
g-prior on fitted values, n_0 in precinct-equivalents	NYC	interpretable, decays correctly, exact conjugacy
Anytime-valid confidence sequence	NYC	the only honest interval under continuous monitoring
Draw-wise turnout-weighted aggregation	NYC	propagates both uncertainties jointly
Two-gate reporting logic (`obs` / `obs_turn`)	NYC	separates "votes in" from "count complete"
Vote mode as a dimension of the unit	TX	the composition mechanism is structural
Strata + partial pooling	TX	required once districts are small
Small, fixed covariate set	TX	the only thing that scales to hundreds of races
Post-hoc normalisation of independent share fits	*neither*	deflates predictive variance; induces rather than estimates correlation
Conformal residual quantiles	*neither*	assumes perfect cross-precinct correlation
Hardcoded λ	*neither*	effective strength varies with data scale

The trial model in `tx_gov_trial.R` (documented separately in `tx_gov_model.tex`) is an attempt at the top half of that table: a Gaussian, conjugate, g-prior system in which vote mode is a dimension of the unit and the composition is factored rather than normalised after the fact.

9. Provenance of the diagnostics

- NYC run log: `nyc/reports/data-out/nyc_small_history.json`, 24 records 2025-11-05 02:31–05:27 UTC plus one at 19:48.
- NYC final snapshot and interval widths: `nyc_current.json`.
- NYC projection losses: centred R^2 of each `pred_*` column on each spec's design, computed on `nyc/data/nyc/nyc_pre.parquet` (4,061 rows).
- NYC estimator internals: `forecast_prec`, `lm_bayes`, `sim_fitted`, `cred_int`, `conf_seq`, `fit_t`, read from the installed `enightmodels` lazy-load database.
- TX run log and interval widths: `tx-prim-26/data-out/log_preds.ndjson`, 21 records 2026-03-04 01:38–21:14. Confirmed real data (`src = 0`): the logged times match none of the 31 hardcoded `fake_times` on 2026-03-03.
- TX stage distribution: `tx-prim-26/data-out/precincts.csv`.
- `glmnet` shrinkage: Poisson fit on Cornyn counts from the same file.

One caveat on the NYC interval diagnostics: `fit_t` computes `MASS::fitdistr(x, "t")` and then discards the result, always returning (mean, sd, df = ∞). Both NYC intervals are therefore Gaussian approximations rather than the t-fits the function name implies, and `cred_int` measures roughly 283× slower than a plain quantile call entirely on discarded computation. `cred_int` also has no $[0, 1]$ clamp, so a small candidate's lower bound can print negative, though it did not at the final snapshot.